

Chapter 162 — Radiation Therapy for Prostate Cancer

RADIATION THERAPY BACKGROUND AND TECHNOLOGIC ADVANCES

Early Radiation Treatment With External Beam and Brachytherapy Approaches

- **Roentgen described the x-ray in 1896** ("Concerning a New Kind of Ray") and received the **first Nobel Prize in Physics in 1901**. The mechanism of tissue damage — **ionisation causing deoxyribonucleic acid (DNA) damage** — was identified only decades later, after DNA itself had been described.
- **First prostate radiotherapy: Pasteau and Degrais, 1913 (published 1914)**, using an intraurethral radium source. **Hugh Hampton Young**, pioneer of the radical perineal prostatectomy, adopted and popularised it.
- **Kilovoltage external beam radiotherapy (EBRT), 1930s–1950s**: these low-energy beams deposit their dose in skin and superficial soft tissue, so an effective prostate dose could not be reached without unacceptable toxicity.
- **Three requirements for treating a deep pelvic target**. These drove every later advance:
 1. **Penetrating beams** that deposit little dose along their path, sparing skin and tissue.
 2. **Sparing of adjacent organs**: rectum, bladder and femoral heads.
 3. **Minimal targeting error** from physiological organ motion and tissue deformation between visits.

Linear Accelerators

- **1960s: megavoltage EBRT** made curative treatment possible, with acceptable normal-tissue damage and less skin toxicity.
- **Mechanism**: electrons are accelerated along a linear path of oscillating electric potentials → strike a target (tungsten) → **photons (x-rays)** are produced → the beam is shaped by lead blocks, tungsten collimators and leaves.
- **Higher photon energy → deeper dose deposition and greater skin sparing**. Orthovoltage (100–500 kV) and cobalt-60 (1.17–1.33 MV) units deposit their maximum dose at or just below the skin surface. Modern linear accelerators (**6–20 MV**) place the maximum dose **several centimetres deep**.
- **1980s–1990s: computer-based planning** → CT-based three-dimensional conformal radiotherapy (3D-CRT), then intensity-modulated radiation therapy (IMRT).

CT-Based Treatment Planning and Three-Dimensional Conformal Radiotherapy

- **Conventional radiation (until the mid-1980s)**: fields were designed from surface and skeletal landmarks. The prostate was localised on plain films using a contrast-filled Foley catheter or urethrogram and a rectal tube, with portals centred on the pubic symphysis and femoral heads.
- **1990s: computed tomography (CT) entered radiation oncology**. The prostate could now be seen at planning, and computing power allowed dose calculation corrected for the differing **electron densities** of bone, air and soft tissue.
- **Planning workflow: simulation** (a CT scan in the immobilisation device used for daily treatment) → the radiation oncologist contours the target (prostate) and the normal tissues (**regions of interest, ROIs**) → prescribes the target dose, its acceptable variation and the normal-tissue constraints → dosimetrists and physicists calculate and optimise → **dose-volume histograms (DVHs)** record the dose each structure receives.
- **3D-CRT**: the beams conform to the target's shape. Its purpose is to **improve the therapeutic index**, escalating prostate dose while respecting adjacent organs. Planning without CT data produced frequent errors in field placement and portal size.

Basics of Fractionation in Radiation Therapy

- **Coutard (1932)**: many small fractions give disease control with acceptable normal-tissue tolerance.
- **Conventional fractionation = 1.8–2.0 gray (Gy) daily**. One gray is the absorption of 1 joule per kilogram. This fraction size rests on early clinical experience, animal data and mathematical models.
- **Linear-quadratic model**: the most widely used model for predicting a tissue's response to fraction size, based on the tissue's α/β ratio.
- **Why small fractions were preferred**: late-responding normal tissues were thought more sensitive to fraction size than tumour. **Late toxicity (≥ 90 days after treatment)** is reduced by fractionation, so more fractions spare late-responding tissue more than tumour.

A/B RATIO — THE GOVERNING CONTRAST

Feature	Small α/β (2–4 Gy)	Large α/β (≥ 8 Gy)
Typical tissue	Late-responding normal tissue (late sequelae)	Most tumours (early-responding)
Sensitivity to fraction size	High	Low

Radiobiologic Basis for Hypofractionation

- Most models assume tumours have more cycling cells than late-responding normal tissue. **Prostate cancer has relatively few cycling cells**, so it may respond to fraction size and dose rate **like a late-responding normal tissue**.
- Classic α/β estimates are **~ 3 Gy for normal tissue and ~ 10 Gy for tumour**. **Prostate cancer's α/β is estimated as low as ~ 1.5 Gy**, lower than that of the surrounding late-responding organs.
- **Consequence**: using fractions larger than 2 Gy allows a lower total dose, which means less normal-organ injury with the same or better cancer control.
- **Evidence**: the first modelling study pooled brachytherapy and EBRT outcomes and found $\alpha/\beta \approx 1.5$ Gy best fitted the data. It was criticised for mixing modalities that differ in both dose per fraction and spatial dose distribution. Larger EBRT-only cohorts then confirmed **$\alpha/\beta \approx 1.0$ – 1.7 Gy** (varying with risk group) and ≈ 1.4 Gy.

Concept — Prostate cancer's low α/β reverses the usual fractionation logic. Because the tumour, not the rectum, is the tissue most sensitive to fraction size, **larger fractions favour tumour kill over late toxicity**, and every hypofractionation and stereotactic regimen later in the chapter follows from this.

Transrectal Ultrasound-Guided Brachytherapy

- **Brachytherapy (BT)**, from the Greek *brakhus* ("short"): radioactive sources placed in or near a tumour.
- **Historical sequence**: Pasteau placed temporary radium needles through the urethra (1914) → **Young** used intracavitary radium in the bladder, rectum and urethra, "cross-firing" into the prostate (1922) → **Barringer** (Memorial Hospital, New York) implanted radon needles transperineally → **Whitmore** performed retropubic ^{125}I implantation with lymph node dissection (1970s).
- **Retropubic "freehand" technique**: open exposure; needles passed anterior to posterior, guided by a gloved finger in the rectum. Complications: **bleeding from the periprostatic venous plexus**, needles passing through the rectum or bladder, and infection.
- **Holm: transrectal ultrasound (TRUS)-guided transperineal technique**, giving live image guidance. Set-up: high lithotomy → TRUS probe held on a **stepper** fixed to the table or a floor pedestal → a predrilled **template** with parallel holes sits on the perineum, its grid overlaid on the ultrasound image → needles inserted through the template → sealed seeds deployed **as the needle is withdrawn**.
- Trans-sacral CT-guided and intraoperative magnetic resonance imaging (MRI)-guided implantation have been studied but are **not in routine use**.

- *Fig. 162.1 — fluoroscopy: (A) rectal ultrasound probe and BT needles in place; (B) seeds deployed in the prostate after needle removal.*

KEY POINTS: EARLY RADIATION TREATMENT WITH BEAM AND BRACHYTHERAPY APPROACHES

- **Higher photon energy** → more skin sparing, deeper dose. **DVHs** summarise dose to each structure. **LDR BT is TRUS-guided and peripherally loaded** to spare the urethra.

Concept — Every technical step in this history (megavoltage, CT planning, conformal shaping, TRUS guidance) answers one of three problems: **penetration, organ sparing and targeting accuracy**. Fractionation adds a separate, biological way of controlling toxicity.

CONTEMPORARY TECHNIQUES AND ADVANCES IN IMAGING

Multiparametric Magnetic Resonance Imaging in Radiation Treatment Planning

- **Multiparametric MRI (mpMRI)** gives anatomical and functional information at diagnosis (see Chapter 152) and upgrades or upstages disease. It predicts **seminal vesicle invasion more accurately than extracapsular extension**.
- **Limitation:** against whole-mount pathology in the Prostate Imaging–Reporting and Data System version 2 (PI-RADS v2) era, MRI still misses some clinically significant cancers, particularly those **<10 mm**.
- So the workup before definitive radiotherapy uses **systematic ± targeted biopsies**. **Several cores from the same target count as one core** when calculating the proportion of positive cores (National Comprehensive Cancer Network, NCCN).
- **MRI simulation:** MRI carries **no electron-density information**, so it cannot calculate dose on its own. It is **fused with the CT simulation** to delineate the prostate and tumour.
- **MRI is most useful when CT is compromised**, for example by a **hip prosthesis** or a **small prostate**. Some centres also use it in post-implant quality assurance for LDR seed distribution.
- **Erectile vessel-sparing radiotherapy**, planned on T2 sequences and MR angiography: a prospective single-arm study found most men still sexually active at 5 years. The radiosensitivity of neurovascular structures is poorly quantified, so the technique remains investigational.

Metabolic Imaging in Prostate Cancer

- **Prostate-specific membrane antigen (PSMA) positron emission tomography (PET)/CT is appropriate before treatment for unfavourable intermediate-risk and high-risk intact disease** (see Chapter 152). Prospective trials using ^{68}Ga -PSMA or ^{18}F -DCFPyL show that **up to about half** of these patients have their treatment recommendation changed.
- **After prostatectomy**, ^{68}Ga -PSMA, ^{18}F -DCFPyL, ^{18}F -fluciclovine and ^{18}F -choline all detect microscopic disease better than conventional imaging.

Intensity-Modulated Radiation Therapy

- **IMRT** intensifies dose to the prostate while reducing dose to the surrounding tissues. It evolved from 3D-CRT.
- **Evolution:** static beam angles with moving shields in the linear accelerator head → **inverse planning**, in which the user enters the prescription, normal-tissue tolerances and structure priorities → the optimiser generates a candidate plan → clinical review.
- IMRT produces **steep dose gradients at the edge of the prostate**. *Fig. 162.2 — IMRT plan, 79.2 Gy (7920 cGy) in 44 fractions to the prostate, sparing most of the rectum and bladder.*

- Planning studies show IMRT **saves the rectum, bladder, femoral heads and small bowel** better than 3D-CRT. **No randomised trial has compared dose-escalated IMRT with 3D-CRT**, yet almost all prostate radiotherapy in the United States was IMRT by 2009.
- **Volumetric modulated arc therapy (VMAT)**: a subtype of IMRT that treats continuously during a **360° gantry arc**. Treatment time is about halved.

	3D-CRT	IMRT / VMAT
Planning	Forward : beams shaped to the target outline	Inverse : constraints entered, optimiser computes the fluence
Dose shaping	Conforms to the target's shape	Modulated intensity; steep gradients; concave shapes around the rectum

Image-Guided Radiation Therapy and Treatment Margins

Interfraction motion	Intrafraction motion
Prostate lies in a different position on different days	Prostate moves or deforms during a single session , while the beam is on
Counteracted by daily online imaging before treatment	Counteracted by real-time tracking, endorectal balloon or transponders

INTERNATIONAL COMMISSION ON RADIATION UNITS (ICRU) TARGET VOLUMES

Volume	Definition	Prostate application
Gross tumour volume (GTV)	Clinically or radiographically evident tumour	Dominant intraprostatic lesion (DIL) on diagnostic or planning MRI (NRG Oncology)
Clinical target volume (CTV)	GTV + areas of suspected microscopic disease	GTV + whole prostate ± proximal seminal vesicles (depending on the risk of microscopic seminal vesicle invasion) ± elective pelvic nodes : distal common iliac, external iliac, internal iliac, obturator, presacral
Planning target volume (PTV)	CTV + margin for organ motion, set-up error and beam inaccuracy	Margin of 3–7 mm with modern image guidance

- **The PTV trade-off**: a larger margin makes target coverage more certain but irradiates more normal tissue. Image guidance confirms the target's position moments before treatment, so margins have shrunk steadily.
- **Online image-guided radiation therapy (IGRT)** means daily imaging to confirm position immediately before delivery, using implanted fiducial markers, cone-beam CT or three-dimensional ultrasound.

CT-BASED IGRT SYSTEMS

System	Configuration
Helical kilovoltage on-rail CT	A mobile scanner separate from the linear accelerator
Kilovoltage cone-beam CT	Flat-panel detectors and a kilovoltage source integrated into the linear accelerator
Helical megavoltage CT	The treatment beam itself is aligned to a detector in the machine head

- All three can align to **fiducials, soft tissue (prostate, bowel, bladder) or bone** (Fig. 162.3 — (A) linear accelerator with imaging and treatment head together; (B) simulation images overlaid on online images taken just before the beam turns on; (C) a rotated pelvis misaligns the fiducials, so the table must be moved or the patient repositioned before treatment).
- **Benefit**: better biochemical control, **less genitourinary (GU) bother and less radiation proctitis**. **Cost**: a small extra integral dose, about **2–3 cGy per scan**.
- **Transabdominal ultrasound** is no longer used because of **interuser variability**.

- **Intrafraction control:** an **endorectal balloon** fixes the prostate against the pubic symphysis, allowing tighter margins (*Fig. 162.4* — sagittal plan with rectal balloon). **Radiofrequency transponders** give 3D localisation in the treatment room but cause **MRI artefact**. **Bladder-filling and rectal-emptying instructions** reduce both inter- and intrafraction uncertainty.
- **Image guidance is the contemporary standard** (NCCN), and it made hypofractionation and stereotactic treatment possible.

Stereotactic Body Radiation Therapy

FRACTIONATION CATEGORIES

Category	Dose per fraction	Typical course
Conventional	1.8–2 Gy	~8 weeks (e.g. 78 Gy in 39)
Moderate hypofractionation	2.25–4 Gy	~4 weeks (e.g. 60 Gy in 20)
Ultrahypofractionation (extreme hypofractionation) = stereotactic body radiation therapy (SBRT) or stereotactic ablative radiotherapy (SABR)	≥5 Gy (5–10 Gy)	≤5–7 fractions (e.g. 36.25 Gy in 5)

PHASE III TRIALS OF ULTRAHYPOFRACTIONATION AND SBRT

Trial	Population	Comparison	Result
HYPO-RT-PC (Scandinavian)	Intermediate–high risk (11% high risk); no ADT	42.7 Gy in 7 (6.1 Gy) vs 78 Gy in 39	Noninferior failure-free survival (5-year ~84% in both arms). More acute toxicity early; late toxicity equal
PACE-B	Low–intermediate risk	36.25 Gy in 5 vs 78 Gy in 39 or 62 Gy in 20	2-year late grade ≥2 GU (2% vs 3%) and gastrointestinal (GI) (3% vs 2%) toxicity similar ; biochemical follow-up ongoing
PACE-A	Localised disease	Radical prostatectomy (RP) vs SBRT	Final publication pending (abstract only)

- **Dose ceiling:** prostate-specific antigen (PSA) nadirs are deeper at 45–50 Gy in 5 fractions, but a meta-analysis of prospective studies found **no oncological gain above 40 Gy in 5 fractions (or 36 Gy in 4)**, with possibly more late toxicity.
- **Delivery: every other day** (two to three times a week) to allow normal-tissue repair, so five fractions fit within about 2 weeks. Requires **stringent image guidance and margins of 3–5 mm**, usually with intermittent real-time fiducial tracking or on-board CT.
- SBRT plans are **heterogeneous**, with higher doses within parts of the gland.
- **Platforms:** a standard linear accelerator; **CyberKnife** (an X-band linear accelerator on a robotic arm), which **needs intraprostatic fiducials** for intrafraction tracking, takes longer and cannot show bladder or rectal filling; and the **MRI-linear accelerator**.
- **Utilisation:** SBRT use in the United States rose steadily from a negligible share in 2004, spreading from academic centres into the community. The **commonest schedule is five fractions: 725 cGy × 5 (36.25 Gy)**, then 700 cGy × 5 (35 Gy).

Magnetic Resonance Imaging in Radiation Treatment Delivery

Trial	Question	Result
FLAME (phase III)	Focal boost to the MRI-defined DIL, intermediate- and high-risk disease	Better biochemical progression-free survival (hazard ratio [HR] 0.45, $p < 0.001$), with no worse toxicity or quality of life
DELINEATE	DIL boost vs homogeneous whole-gland IMRT	Acceptable toxicity and patient-reported outcomes
hypo-FLAME / hypo-FLAME II	DIL boost within 5-fraction SBRT	No grade ≥ 3 toxicity. Nonrandomised comparison: once-weekly gave fewer acute toxicities than twice-weekly
MIRAGE (phase III, 2020–2021)	MRI-guided SBRT vs CT-guided SBRT	Margins 2 mm vs 4 mm. Less acute grade ≥ 2 GU toxicity (physician-reported, Radiation Therapy Oncology Group [RTOG] scale). Patient-reported incontinence (Expanded Prostate Cancer Index Composite [EPIC]-26) worse at 1 month with CT guidance, equal by 3 months , which has been criticised. Primary endpoint positive; long-term data awaited

- *Fig. 162.5* — deliberate DIL boost to 67 Gy with the whole prostate at 60 Gy, integrated in 20 fractions.
- **Advantages of the MRI-linear accelerator:** **better soft-tissue resolution** than CT-based machines; **no fiducial markers** needed; beam arrangements **adapt in real time** to daily prostate deformation and bladder and rectal shape; real-time target tracking. MRI-guided radiotherapy (MRgRT) began with conventional and moderately hypofractionated regimens, and SBRT programmes have grown over the past 5 years.
- **Disadvantages:** online adaptive optimisation is **time-consuming**, so fewer patients can be treated per day, and the imaging delay itself **allows further internal deformation**.

Heavy Particle Beam/Proton Therapy

- **Particles in clinical use:** electrons (shallow penetration → **skin and superficial tumours**); protons; neutrons; helium ions; heavy ions (neon, argon, carbon); negative ions. Heavy particles need large infrastructure and are hard to control.
- **Physical advantages:** a higher **linear energy transfer (LET)**, the energy deposited per unit path length → denser DNA damage and less repair → a slightly higher **relative biological effectiveness (RBE)** than photons.
- **Bragg peak:** a sharp rise in dose, then a cut-off at the end of the particle's range. Tumours are thicker than a single peak, so the beam is modulated into a **spread-out Bragg peak (SOBP)**.
- *Fig. 162.6* — dose against depth: photons deliver a surface dose and an exit dose; protons have a lower entrance dose, a Bragg peak and no exit dose, and the SOBP covers the tumour depth. (From Leeman et al., *Lancet Oncol*, 2017.)
- **Uncertainties:** **range uncertainty** is far more sensitive to set-up, organ motion and tissue heterogeneity than with photons, and **RBE varies at the edges of the SOBP**. Either can underdose the target or overdose normal tissue.
- Protons are the most available heavy particle (produced by a linear accelerator, cyclotron or synchrotron). Neutrons were historically more common, and helium and carbon ions are used in a few countries outside the United States.

RANDOMISED TRIALS OF PARTICLE THERAPY

Trial	Comparison	Result and caveat
RTOG neutron trial (1977)	Neutrons vs photons	10-year survival favoured neutrons, but the arms were imbalanced . It still led the National Cancer Institute (NCI) to build neutron facilities
RTOG neutron trial (mid-1980s)	Neutrons vs photons	Better local control and freedom from PSA failure with neutrons, but more severe morbidity and no survival difference
Proton boost trial (1980s), locally advanced disease	Proton vs photon boost	Better 5-year local control with protons, at the cost of more grade 1–2 rectal bleeding and urethral strictures ; no grade 3–5 events. Rectal dose–volume analysis produced benchmark guidelines
PROG 95-09 (Proton Radiation Oncology Group), low–intermediate risk	50.4 Gy photons + proton boost to 70.2 vs 79.2 Gy-equivalent	Higher dose reduced 10-year biochemical failure (~32% vs ~17%), with no difference in overall survival or late toxicity . A dose-escalation trial: no photon comparator , so it says nothing about protons vs photons

TABLE 162.1 TREATMENT PLANNING STUDIES COMPARING INTENSITY-MODULATED RADIATION THERAPY VERSUS CONFORMAL PROTON THERAPY

Dose region	Rectum	Bladder	Femoral head	Small bowel
Low dose (V10–V30) and mean dose	Lower with protons	Lower with protons	Higher with protons (V30 and mean dose)	Lower with protons
Intermediate dose (V50)	Lower or similar with protons	Similar	—	—
High dose (V70–V78)	Similar (small or no difference)	Similar or higher with protons	—	—

Targets: prostate ± seminal vesicles ± lymph nodes. Data from Chera et al., 2009; Vargas et al., 2008; Trofimov et al., 2007.

- **What the table shows:** protons shrink the **low-dose bath** but **not the high-dose volume** of rectum and bladder, which is what drives late toxicity. Angling the beams away from lateral paths did not change this.
- **The trap:** protons are not better for every organ. **Opposed lateral proton beams** (used classically so that range uncertainty runs parallel to the prostate–rectum interface) pass through the femoral heads. **IMRT gives similar or lower femoral head dose**, especially when pelvic nodes are treated, and femoral head dose is linked to **late avascular necrosis**.
- **Clinical comparisons:** a case-matched study showed **no difference in grade ≥2 acute or late GI or GU toxicity**. A matched Surveillance, Epidemiology, and End Results (SEER) analysis showed **no reduction in second malignancies** with protons.
- Spot-scanning improves dose shaping, but **no trial compares protons with contemporary IMRT**. The evidence is single-arm series with high biochemical control by risk group; urinary irritative and bowel scores fall at 1 year and **return almost to baseline over 3–4 years**. Outcomes appear **similar to IMRT**, and cost with no measurable benefit limits wider use.
- **PARTIQoL** (ongoing randomised trial): protons vs IMRT in low- or intermediate-risk disease; primary endpoint **mean EPIC bowel score at 24 months**.

KEY POINTS: CONTEMPORARY TECHNIQUES — EXTERNAL BEAM RADIATION THERAPY

- **Margins now subcentimetre; IMRT the commonest curative method; hypofractionation routine; ultrahypofractionation a standard option for all risk groups;** protons have higher RBE and need more infrastructure, with **no randomised comparison with IMRT.**

Concept — Modern EBRT is a **margin-reduction chain**: imaging (MRI, PSMA) defines the target, IMRT shapes the dose, and IGRT or MRgRT shrinks the margin. Only smaller margins make large fractions safe. Protons improve the physics of the low-dose region, but toxicity comes from the high-dose region, which is why no clinical benefit has been shown.

CONTEMPORARY TECHNIQUES: BRACHYTHERAPY**Low-Dose-Rate Brachytherapy Technique and Trends**

- **LDR BT:** sources are implanted **permanently** and deliver dose gradually until they have fully decayed. "Low dose rate" describes the **rate**, not a low dose.
- **Low photon energy (~21–30 kiloelectron-volts, keV) → dose falls off within a few millimetres.** This spares surrounding tissue but demands precise placement, or tumour-bearing regions are underdosed.
- **Utilisation:** BT for low-risk disease rose through the 1990s (Cancer of the Prostate Strategic Urologic Research Endeavor [CaPSURE] database), then **fell steeply** (by about 60%). Proposed causes: competition from **ultrahypofractionated EBRT**, **robotic prostatectomy**, and urology practices **owning linear accelerators**. Evidence that adding BT to EBRT improves control and survival in intermediate- to high-risk disease may reverse the decline.
- **Planning:** computer algorithms arrange the sources to maximise prostate ablation while limiting dose to the **intraprostatic urethra and rectum** (Fig. 162.7 — (A) preoperative MRI-based plan; (B) intraoperative ultrasound-based plan).
- **Peripheral (modified peripheral) loading:** more seeds at the edge of the gland and fewer centrally → **urethral sparing** → less urinary morbidity.

IMPLANT PLANNING METHODS

Method	Basis
Preplanned (Seattle; Blasko)	Preoperative TRUS images taken in the same lithotomy position as the implant
Nomogram (Stone and Stock)	Seed number from gland size using the Anderson nomogram; placement by the Paterson–Parker empirical rules (1930s)
Intraoperative planning (Kaplan)	Dosimetry calculated on TRUS images acquired during the implant

- A comparison of the three methods found that **intraoperative planning gave the best prostate coverage and the lowest mean and maximum urethral doses.**
- **Cystoscopy during the implant** is used to irrigate bladder clots or, rarely, to retrieve seeds placed in the bladder.

LDR ISOTOPES

Feature	¹²⁵ Iodine	¹⁰³ Palladium	¹³¹ Caesium
Photon energy	~27–28 keV	~21 keV	~29–30 keV
Half-life	59.6 days	17 days	9.7 days
90% of dose delivered by	204 days	58 days	33 days

Feature	¹²⁵ Iodine	¹⁰³ Palladium	¹³¹ Caesium
Introduced	Earliest of the three	1986	2004
Monotherapy dose	140–160 Gy	110–125 Gy	100–115 Gy
Boost dose (with 45 Gy EBRT)	108–110 Gy	90–100 Gy	80–85 Gy

- **Pattern:** shorter half-life → faster delivery → lower total prescription. ¹²⁵I is called "**more forgiving**" because its slightly higher energy keeps coverage good through the implant and the months of decay.
- **Randomised trial of ¹²⁵I vs ¹⁰³Pd** (low-risk disease: ≤cT2a, Gleason ≤6, PSA ≤10 ng/mL): **no difference in biochemical control**. ¹⁰³Pd gave **worse urinary symptoms at 1 month** (American Urological Association [AUA] symptom score) but **faster resolution by 6 months**, matching its shorter half-life.
- **Low risk → BT monotherapy** gives excellent results. **Unfavourable intermediate or high risk → BT + EBRT:** EBRT raises the dose to periprostatic tissue and seminal vesicles, can include elective pelvic nodes and compensates for underdosed parts of the gland. Together they reach a higher cumulative dose than either can safely give alone.
- **Combined regimen:** EBRT **45 Gy** (compared with 75–80 Gy for EBRT alone) + an implant at **60–70% of the monotherapy dose**.

Assessment of Permanent Implant Quality

- **Post-implant imaging at 14–30 days** (isotope-dependent), by CT and/or MRI: seed count and arrangement are checked to confirm a tumoricidal prostate dose with minimal dose to adjacent structures (Fig. 162.8 — post-implant dosimetry: isodose lines at 50–200% of prescription and the DVH for prostate, urethra, rectum and bladder).
- **D90** = the **minimum dose received by the "hottest" 90%** of the prostate (the highest isodose line covering 90% of the gland). **V100** = the **percentage of the prostate receiving the prescribed dose**. The American Brachytherapy Society (ABS) recommends **D90, V100 and V150**.

TABLE 162.2 POSTIMPLANT DOSIMETRY METRICS

Organ	Parameter	Constraint
Prostate	D90: dose to 90% of the prostate	>90% of prescription dose
	V100: volume receiving 100% of prescription	>85% of the prostate
Urethra	Volume receiving 150% of prescription	<0.05 cm ³
	Volume receiving 115% of prescription	<0.1 cm ³
Rectum	Volume receiving 100% of prescription (RV100)	<1 cm³ for ¹²⁵ I/ ¹⁰³ Pd; <0.5 cm³ for ¹³¹ Cs

- **Purpose:** coverage (D90, V100) is set as a floor, and dose to the urethra and rectum as a ceiling. The rectal limit is **tighter for ¹³¹Cs**.
- **Trap:** D90 is a **dose** (as a percentage of prescription); V100 is a **volume** (percentage of the gland).
- **Sources of error between plan and result:** needle deviation, tissue deformation, **prostate oedema** and seed shifts after needle withdrawal.
- **Oedema:** the gland swells substantially and variably on day 1 and settles over **2–4 weeks**. With a short half-life, much of the dose is delivered before the swelling resolves, which is why the timing of post-implant imaging varies (day 0–30).
- **Dosimetry predicts outcome:** **lower prostate D90**, higher Gleason score and higher initial PSA each predict biochemical failure. A multi-institutional analysis confirmed the D90–outcome link for both ¹²⁵I and ¹⁰³Pd.

Real-time (ultrasound) dosimetry	Dynamic dosimetry
Source positions updated during the implant from the needle tip or source seen on sagittal TRUS ; commercially available and widely used	Sources seen directly on repeated fluoroscopy or in-room CT during the implant, e.g. C-arm cone-beam CT co-registered with TRUS via gold fiducials, or a nonisocentric fluoroscopy unit (three images → 3D seed map matched to TRUS)
Cannot capture late source movement or oedema; compared with post-implant CT, it underestimates high-dose regions and normal-tissue dose	Allows remedial seeds during the procedure; a phase II trial achieved prostate V100 ≥95% in almost every patient

High-Dose-Rate Brachytherapy

- **High-dose-rate (HDR) BT:** temporary catheters (**hollow transperineal needles**) → imaged by TRUS, CT or MRI → a single high-activity **iridium-192 (^{192}Ir)** source (gamma, mean **380 keV**) in a **remote afterloader** steps through set dwell positions → treatment over minutes → the source is withdrawn.
- **Conformality comes from varying dwell time and position.** Robotic source control protects staff. Because dose is calibrated in real time and the source is removed, **no post-implant dosimetry is needed.**
- **Evolution:** first used as a **boost with EBRT** in intermediate- to high-risk disease; now given as **multifraction monotherapy** for more favourable disease.

HDR COMPARED WITH LDR BRACHYTHERAPY

Advantages of HDR	Disadvantages of HDR
Can include extracapsular disease and seminal vesicles	Catheter migration if the patient is moved for imaging → inaccuracy
Finer optimisation by editing dwell times	Contouring inaccuracy and interobserver variability with periprocedural oedema
Possible biological advantage of rapid delivery given prostate cancer's low α/β	^{192}Ir needs a shielded room
Remote afterloading → no manual handling of radioactive material	Less favourable pathological response on post-treatment biopsy studies
One source serves many patients → may be cost-effective in high-volume centres	

High Dose Rate as a Boost With External Beam Radiation Therapy

EBRT component	HDR boost component
45 Gy in 25 fractions (5 weeks)	12–15 Gy in 1 fraction
46 Gy in 23 fractions (4.5 weeks)	11–22 Gy in 2 fractions
35.7 Gy in 13 fractions (2.5 weeks)	15 Gy in 3 fractions
37.5 Gy in 15 fractions (3 weeks)	

High-Dose-Rate Monotherapy

- **Established schedules:** **27–28 Gy in 2; 31.5 Gy in 3; 34–38 Gy in 4 fractions.** They give excellent biochemical control with limited toxicity.
- **Single-fraction 19 Gy fails.** A prospective series gave disappointing disease-free survival and substantial biopsy-proven local failure at about 4 years. In the randomised comparison of **19 Gy × 1 vs 27 Gy in 2** (median follow-up 60 months), 5-year biochemical disease-free survival was **73.5% vs 95%** and 5-year local failure **29% vs 3%**. **Multifraction HDR monotherapy remains the standard** for low- and intermediate-risk disease.

- **The only randomised LDR vs HDR monotherapy trial** used HDR 19 Gy $\times$ 1. LDR gave a **deeper PSA nadir** (<0.4 ng/mL) at the cost of **more urinary irritative change** (International Prostate Symptom Score [IPSS], EPIC). Final outcomes are pending.

Patient Selection for Brachytherapy

- **The main acute toxicity is obstructive and irritative uropathy**, which leads to a temporary catheter in a small proportion of patients. This is why urinary function drives eligibility.
- **Guidelines:** ABS and Groupe Européen de Curiethérapie–European Society for Radiotherapy and Oncology (GEC-ESTRO) consensus guidelines for LDR and HDR.
- **Prerequisites:** the patient must be **fit for general anaesthesia** (the standard; local anaesthesia appears only in limited reports) and **able to undergo TRUS**, because the technique depends on strict spatial accuracy and TRUS visualisation.

RELATIVE CONTRAINDICATIONS TO BRACHYTHERAPY

- **Large TURP defect, or TURP within the previous 3–6 months** → incontinence risk and technical difficulty.
- **Poor urinary function or obstruction:** maximum flow rate (Qmax) <10 mL/s; IPSS >20 ; large median lobe.
- **Prior pelvic radiotherapy.**
- **Connective tissue disease or inflammatory bowel disease.**
- **Pubic arch interference:** needles cannot reach the anterior or lateral prostate.
- **Lithotomy position not possible.**
- **Rectal fistula.**
- **Cytoreduction of a large gland or median lobe:** a short course of ADT or androgen receptor blockade shrinks the gland by **up to ~30%**, but carries toxicity including **increased cardiovascular mortality**. 5 α -reductase inhibitors shrink it by only ~20% but are a **less morbid alternative**.
- **ADT with BT:** usual in high-risk disease treated with BT + EBRT; its role is **unclear in unfavourable intermediate risk, or high risk with few risk factors**.

Brachytherapy Combined With External Beam Radiotherapy

- The combined regimen is **45 Gy of conventionally fractionated EBRT + BT at ~60–70% of the monotherapy dose**. EBRT covers periprostatic tissue and raises the gland dose.
- A pooled analysis of two randomised trials showed that biochemical failure **rose with risk group** (favourable intermediate → unfavourable intermediate → high) with **no measurable benefit from adding EBRT to BT**.
- With ADT, biochemical disease-free survival after combined therapy in high-risk disease is **80–90%**. HDR series with ≥ 5 years of follow-up show control of **83–100% in intermediate risk and 74–91% in high risk**.

TRIALS OF BRACHYTHERAPY WITH AND WITHOUT EXTERNAL BEAM RADIOTHERAPY

Trial	Population	Comparison	Verdict
RTOG 0232	Intermediate risk	BT alone vs EBRT 45 Gy + BT	No difference in biochemical progression-free survival (5-year 85.5% vs 88%), including in the Gleason 4+3 subgroup (Grade Group 3). More late toxicity with the combination → BT alone is a standard for intermediate risk

Trial	Population	Comparison	Verdict
ASCENDE-RT	High risk (and intermediate)	12 months ADT + pelvic EBRT 46 Gy → EBRT boost to 78 Gy vs LDR boost	EBRT boost had twice the biochemical failure (HR 2.04, P = 0.004) ; no survival difference; benefit sustained at 10 years
United Kingdom HDR boost trial	Localised disease, most with ADT	EBRT 55 Gy ± HDR 17 Gy in 2 fractions	21% absolute relapse-free survival benefit at 12 years (p = 0.008)
Gleason 9–10 pooled analysis (very high risk)	Gleason 9–10, 12 centres	RP vs EBRT vs EBRT + BT	EBRT + BT: lowest 5-year prostate cancer mortality (3% vs 12% RP vs 13% EBRT) and fewer distant metastases

RTOG 0232 LATE TOXICITY — TWO REPORTS

Report	EBRT + BT	BT alone	Why the figures differ
Long-term (median 12.1 years; 5- and 10-year cumulative incidence)	Grade ≥2: 42.8% / 48.0%; grade ≥3: 8.2% / 9.8%	Grade ≥2: 25.8% / 30.6%; grade ≥3: 3.8% / 4.2%	Different data cuts (initial 5-year report vs long-term cumulative incidence) and different endpoint definitions
Initial 5-year report	Late grade ≥2: 53%; grade ≥3: 12%	Late grade ≥2: 37%; grade ≥3: 7%	

KEY POINTS: CONTEMPORARY TECHNIQUES — BRACHYTHERAPY

- **LDR (permanent) or HDR (temporary); implant quality (V100, D90) determines control; IPSS >20 or Qmax <10 mL/s is a relative contraindication; BT alone standard for low and intermediate risk, BT + EBRT for higher risk.**

Concept — BT's advantage is **dose geometry**: the steep fall-off lets the intraprostatic dose far exceed what EBRT can give. Two consequences follow: **outcome depends on implant accuracy (D90)**, and **eligibility depends on whether the urethra can tolerate acute swelling**. EBRT is added only where disease may lie beyond the implant's reach.

RADIATION THERAPY FOR METASTATIC DISEASE

Definitions of the Metastatic State

- Standard staging records **whether** metastases are present, not **how many**. Modern imaging counts lesions precisely, from one to five small lesions up to a "**superscan**" (diffuse bone uptake with no tracer in the kidneys or bladder).

Term	Definition	Role of local therapy
Oligometastatic, de novo	Limited metastases at initial presentation	Eradicating all visible tumour may give durable control or cure
Oligorecurrence	Limited metastases on follow-up after curative surgery or radiation, without systemic therapy	As above
Oligoprogression	Limited progression while on otherwise effective systemic therapy	Treat the resistant sites to preserve the current systemic therapy and avoid switching to a less effective or more toxic one
High volume (CHAARTED)	≥4 bone metastases with ≥1 outside the vertebral bodies or pelvis, and/or visceral metastases	Systemic and radiopharmaceutical therapy

METASTATIC HORMONE-SENSITIVE DISEASE: BURDEN AND TREATMENT OF THE PRIMARY

Trial	Comparison	Result
CHAARTED	ADT + docetaxel vs ADT	Overall survival 57.6 vs 44.0 months (HR 0.61) . No benefit in low-volume disease → low volume behaves differently
HORRAD	ADT ± prostate RT, any number of bone metastases	No overall survival difference ; trend favouring patients with ≤5 metastases
STAMPEDE (metastatic RT comparison)	Standard care (lifelong ADT ± upfront docetaxel) ± prostate RT	Unselected: no overall survival benefit ; failure-free survival improved (HR 0.76). Low burden (CHAARTED definition): overall survival improved (HR 0.68; 3-year 73% → 81%)

Radiopharmaceutical Therapies for High-Volume Disease

- EBRT relieves symptoms, but when disease is extensive not every site can be treated without excess normal-tissue dose. **Systemic radiopharmaceuticals** target bone turnover or PSMA. They can **improve survival, slow progression** and reduce fractures, spinal cord compression and other skeletal events.
- **PSMA** is also called **folate hydrolase I or glutamate carboxypeptidase II**. It is imaged with ^{18}F or ^{68}Ga carriers; the U.S. Food and Drug Administration (FDA)-approved therapeutic radionuclide is ^{177}Lu .

RADIOPHARMACEUTICALS — TARGETS AND MECHANISM

Agent	Target / emission	Key properties	Pivotal evidence
$^{223}\text{Radium}$	"Calcimimetic": binds hydroxyapatite in new bone around osteoblastic metastases; alpha emitter	Half-life 11.4 days ; range <100 µm ; double-strand DNA breaks; eliminated via the GI tract	ALSYMPCA
$^{89}\text{Strontium}$	Calcium analogue taken up by forming bone	Pain reduced in most patients, pain-free in few ; better quality of life	Palliative series
$^{153}\text{Samarium-lexidronam}$	Bone-seeking	Pain relief within 1–2 weeks ; less haematological toxicity than ^{89}Sr	Phase III placebo (^{152}Sm)-controlled trial
$^{177}\text{Lu-PSMA-617}$	PSMA-targeted radioligand	Targets the tumour itself, not bone turnover	VISION

RADIOPHARMACEUTICALS — SIGNATURE TOXICITIES

Agent	Toxicity
^{223}Ra	Mild myelotoxicity ; favourable safety profile
^{89}Sr	Myelosuppression (the major toxicity) → needs adequate marrow reserve
^{153}Sm	Mild, transient marrow suppression; no grade 4 events
$^{177}\text{Lu-PSMA-617}$	Common: fatigue, dry mouth, nausea (mostly grade 1–2). Grade ≥3: mainly haematological (anaemia, thrombocytopenia, lymphopenia)

- **ALSYMPCA** (phase III, placebo-controlled; 2:1): castration-resistant prostate cancer (CRPC) with **no visceral metastases** (only non-bulky nodes **<3 cm** allowed, because the agent is not taken up by cancer cells). Patients were post-docetaxel or unfit for or declining it. **50 kBq/kg intravenously every 4 weeks × 6**. Overall survival **14.9 vs 11.3 months (+2.6 months)**; time to first symptomatic skeletal event **15.6 vs 9.8 months**.
- **VISION** (2:1): **PSMA-positive metastatic CRPC (mCRPC) after taxane chemotherapy**. $^{177}\text{Lu-PSMA-617}$ + best supportive care vs best supportive care, which excluded investigational agents, cytotoxic chemotherapy, other systemic radioisotopes and hemibody radiotherapy. Overall survival **15.3 vs 11.3**

months (HR 0.62); radiographic progression-free survival **8.7 vs 3.4 months (HR 0.40);** grade ≥ 3 adverse events **52.7% vs 38%.**

- **Ongoing: PSMAfore** (PSMA radioligand vs a change of androgen receptor-pathway agent in mCRPC); **PSMAAddition** (upfront PSMA therapy with standard care in metastatic hormone-sensitive disease).

Metastasis-Directed Therapies for Oligometastatic Disease

Trial	Population	Comparison	Endpoint and result
STOMP (phase II)	Oligorecurrent; 1–3 extracranial lesions after curative-intent therapy	Metastasis-directed therapy (MDT: surgery or SBRT) vs surveillance	ADT-free survival 21 vs 13 months (HR 0.60, 80% confidence interval [CI]; P = 0.11). Not significant; hypothesis-generating
SABR-COMET (phase II)	Controlled primary, 1–5 metastases , any histology (a minority prostate)	Standard care $\pm$ SBRT to all metastases (1:2)	5-year overall survival 42.3% vs 17.7% (P = 0.006)
ORIOLE (phase II)	Recurrent hormone-sensitive disease, 1–3 metastases	SBRT to all sites vs observation (2:1)	Progression at 6 months 19% vs 61% (P = 0.005)

- **ESTRO–Advisory Committee on Radiation Oncology Practice (ACROP) Delphi consensus:** MDT may be considered for **up to five lesions**, in de novo oligometastatic hormone-sensitive disease, oligorecurrence or oligoprogression. In **oligoprogressive castration-resistant disease**, give MDT without changing the systemic therapy.

KEY POINTS: RADIATION THERAPY FOR METASTATIC DISEASE

- **Low volume** $\rightarrow$ SBRT and local therapy; **high volume** $\rightarrow$ radiopharmaceuticals, with overall survival benefit.

Concept — In metastatic disease, **burden rather than stage decides the role of radiation**. When all visible disease can be targeted, local treatment of the primary and metastases adds survival. When it cannot, radiation has to be delivered systemically.

ROLE OF ANDROGEN SUPPRESSION COMBINED WITH RADIATION

Background and Potential Mechanisms

- **Huggins and Hodges (1941)** showed that prostate cancer is **androgen-dependent**. This provided the rationale for adding hormonal therapy to RT.
- **From 1967**, hormonal therapy was added to RT for **stage C (T3)** tumours. These were often large, and cytoreductive ADT gave **better geometry for irradiation and less tumour burden**. Its use remained controversial through the 1970s–1980s, when early series were often negative.
- **RTOG 75-06**, an early favourable report: patients with histologically unfavourable tumours given androgen suppression + EBRT had **disease-free and overall survival similar to patients with favourable tumours** who did not receive it.
- **Mechanisms of synergy:**
 - **Apoptosis:** androgen-responsive cells undergo programmed cell death when deprived of androgen. Like radiation-induced death, this is **p53-independent**.
 - **Radiosensitisation:** castration $\rightarrow$ less **Ku70** repair protein $\rightarrow$ impaired **nonhomologous end-joining** $\rightarrow$ more unrepaired double-strand breaks after RT.

- **RT upregulates androgen receptor (AR) signalling** durably. Patients with a post-RT rise in AR-regulated hK2 were about **threefold more likely to fail biochemically**. **Adjuvant ADT after RT works by blocking this AR upregulation.**

Androgen Suppression Therapy and Localized Disease

This section and the three that follow draw on one trial series. It is rebuilt once here from Table 162.3 and the text, with rows grouped by the question each trial asks: RT alone vs RT + ADT in localised and in locally advanced disease, ADT alone vs ADT + RT, and ADT duration.

TABLE 162.3 SELECTED RANDOMIZED CONTROLLED TRIALS INVESTIGATING THE USE OF ANDROGEN SUPPRESSION THERAPY WITH RADIATION

Trial	Population	Arms	Key result
Localised disease: RT alone vs RT + ADT			
TROG 96.01 (Trans-Tasman Radiation Oncology Group)	T2b–T4 N0	RT 66 Gy vs + 3 months vs + 6 months ADT	6 months improved disease-free, metastasis-free, disease-specific and overall survival (10-year overall survival 57.5% → 63.3% → 70.8%). 3 months gave no significant prostate cancer-specific mortality benefit
DFCI 95-096 (Dana-Farber)	T1–T2, mostly Gleason ≥7, median PSA 11	RT 70 Gy ± 6 months ADT	8-year overall survival 61% → 74% ; prostate cancer-specific mortality 12% → 3%. Benefit confined to men with no or minimal comorbidity
RTOG 94-08	T1b–T2b, PSA <20	RT 66.6 Gy ± 4 months combined androgen blockade	10-year overall survival 57% → 62% . Benefit in intermediate risk ; none in low risk
Locally advanced disease: RT alone vs RT + ADT			
RTOG 86-10	Bulky (≥25 cm ²) T2b–T4	RT 65–70 Gy ± 4 months total androgen suppression (RT from month 2)	Better local control, disease-free survival and freedom from metastasis; 10-year overall survival not different ; disease-specific mortality 36% → 23%
EORTC 22863 (European Organisation for Research and Treatment of Cancer)	>90% T3–T4	RT 70 Gy ± 3 years goserelin from the start of RT	Overall survival 39.8% → 58.1% (10-year); disease-specific survival also improved
Locally advanced disease: ADT alone vs ADT + RT			
SPCG-7 (Scandinavian Prostate Cancer Group)	Mostly T3, PSA <70, N0	3 months combined androgen blockade → lifelong flutamide ± RT 70 Gy	Relative risk of prostate cancer death 0.44 ; 10-year mortality 39.4% → 29.6%
NCIC PR.3 / MRC PR07	Mostly T3–T4	Lifelong ADT ± RT 65–69 Gy	Disease-specific survival HR 0.54 ; 7-year overall survival 66% → 74%
Duration of ADT			
RTOG 92-02	T2c–T4	4 months vs 4 months + 2 years goserelin	Overall survival not significant overall. Gleason 8–10 subgroup (hypothesis-generating): overall survival 80% vs 69% , disease-specific survival 90% vs 78%

Trial	Population	Arms	Key result
EORTC 22961	T2c–T4, some N1	6 vs 36 months (6 + 30 months luteinising hormone-releasing hormone [LHRH] agonist)	Noninferiority of short course not shown : 5-year prostate cancer mortality 4.7% vs 3.2% , favouring long course
PCS IV	High risk (T3–T4, PSA >20 or Gleason 8–10)	18 vs 36 months	No survival difference ; 18 months gave better quality of life
DART01/05 GICOR	Intermediate–high risk	4 vs 24 months	5-year overall survival 86% → 95% ; benefit in high risk, not intermediate
RTOG 99-10	84% intermediate risk	RT 70.2 Gy + 4 vs 9 months neoadjuvant/concurrent ADT	No advantage to the longer course
TROG 03.04 RADAR	Locally advanced	6 vs 18 months ADT, with rising RT dose (66 → 70 → 74 Gy → 46 Gy + HDR 19.5 Gy in 3)	Local progression fell with dose (22% → 15% → 13% → 7%) . The benefit of extra ADT fell as dose rose , with none clear after an HDR boost

Table source: TROG 96.01 (Denham et al., 2011); DFCI 95-096 (D'Amico et al., 2008); RTOG 94-08 (Jones et al., 2011); RTOG 86-10 (Roach et al., 2008); EORTC 22863 (Bolla et al., 2010); SPCG-7 (Widmark et al., 2009); PR.3/PRO7 (Warde et al., 2011); EORTC 22961 (Bolla et al., 2009); PCS IV (Nabid et al., 2018); RTOG 99-10 (Pisansky et al., 2013); RTOG 92-02 (Horwitz et al., 2008).

- **What the table tests:** ADT added to RT, and RT added to ADT, **each improves survival** in intermediate- to high-risk disease.
- **Pattern:** short ADT (4–6 months) is enough for intermediate risk; high-risk and locally advanced disease need **long ADT (18–36 months)**. Longer courses help Gleason 8–10 disease but not intermediate-risk disease.
- **Traps:** RTOG 86-10 improved disease-specific mortality **but not overall survival**. PCS IV shows **18 months is as good as 36**. Most of these trials used doses well below modern dose-escalated RT.

Androgen Suppression Therapy and Locally Advanced Disease

- **Meta-analysis of ADT in nonmetastatic disease:** relative reductions in both **disease-specific and all-cause mortality**.

Benefit of Radiation in Addition to Androgen Suppression for Locally Advanced Disease

- SPCG-7 and NCIC PR.3/MRC PR07 (see the ADT trials table above) show that **adding RT to lifelong ADT reduces prostate cancer death and improves overall survival**.
- **STAMPEDE (abiraterone comparison)**, in newly diagnosed locally advanced or metastatic disease, or relapsing disease with poor features: **24 months ADT + abiraterone + prednisolone** vs ADT alone gave a **71% relative improvement in time to treatment failure and a 37% difference in overall survival**. RT from 6–9 months after randomisation was **mandatory for node-negative** and encouraged for node-positive nonmetastatic disease.

Optimal Duration of Androgen Suppression

NATIONAL COMPREHENSIVE CANCER NETWORK (NCCN, 2023) ADT DURATION WITH RADIOTHERAPY

Risk group	With EBRT	With BT + EBRT
Favourable intermediate	None	—
Unfavourable intermediate	4–6 months	—

Risk group	With EBRT	With BT + EBRT
High / very high	24–36 months	12–36 months

- It is unclear how long ADT is needed with **dose-escalated or hypofractionated** RT, because many hypofractionation trials used shorter ADT in high-risk patients. Ongoing: **NRG GU-010** (need for ADT in unfavourable intermediate risk, by genomic risk) and **NRG GU-009** (ADT duration in high risk, by genomic risk).

KEY POINTS: RADIATION THERAPY AND ANDROGEN SUPPRESSION THERAPY

- RT + ADT **improves survival** over either alone. **Unfavourable intermediate: 4–6 months. Locally advanced high risk: 28–36 months beats 4–6 months** with conventional RT. Shorter courses may be considered with dose escalation.

Concept — ADT and RT are **mutually sensitising**: ADT impairs DNA repair, and RT switches on AR signalling that ADT then blocks. The **required duration rises with risk and falls as radiation dose rises**, which is why an HDR boost made extra ADT redundant in RADAR.

ROLE OF PROPHYLACTIC PELVIC NODAL TREATMENT

- Pelvic nodes are a natural route of spread, so **prophylactic nodal RT** should in principle reduce recurrence in intermediate- and high-risk disease.

Trial	Setting	Comparison	Verdict
RTOG 94-13 (four arms)	Intact prostate; node risk >15%	Whole pelvis vs prostate only × neoadjuvant vs adjuvant 4 months total androgen suppression	Early 4-year progression-free survival favoured the whole pelvis (54% vs 47%) with no overall survival difference; the benefit was lost on update
GETUG-01	T1b–T3 N0	Pelvis + prostate vs prostate only (6 months ADT allowed in high risk)	No difference in progression-free or overall survival, toxicity or quality of life
RTOG 0534 (SPPORT)	Postoperative salvage	Prostate bed RT vs + short-term ADT vs + pelvic nodes + short-term ADT	Pelvic RT + short-term ADT : better freedom from progression and fewer distant metastases (median 8.2 years)
RTOG 09-24 (open)	Unfavourable intermediate / favourable high risk	Whole pelvis vs prostate	Pending. Specifies the superior border at L4/L5

- **Retrospective data** (node risk $\geq 15\%$): whole-pelvis RT independently predicted better biochemical disease-free survival, with **more acute GI toxicity** but no difference in GU or late toxicity. With VMAT, bowel and urinary quality of life were the same at 24 months.
- **Why the trials may have been negative**: fields ending at **L5/S1 or S1/S2 undercover the common iliac nodes**. The common iliac station is involved in over half of first pelvic nodal recurrences after prostate-only RT. An L5/S1 border covers fewer than half of these recurrences; extending the field to the common iliacs covers **nearly all**.

KEY POINTS: ROLE OF PROPHYLACTIC NODAL RADIATION

- Intact prostate: **no reduction in recurrence in randomised trials**, and **no worse long-term patient-reported outcomes**.

Concept — The negative nodal trials may reflect **field height rather than biology**. Recurrences cluster at the common iliacs above the old L5/S1 border, so the variable to watch is the superior border (L4/L5), not simply whether nodes are treated.

EVALUATING THE RESPONSE TO RADIATION THERAPY

Prostate-Specific Antigen Follow-up: Definition of Failure

- **Serum PSA** is the accepted surrogate endpoint. Unlike after RP, **residual benign glands keep producing some PSA after RT**, but the "acceptable" level is far below age-matched normal because non-malignant acini atrophy.
- **Nadir depends on dose**: median nadir ~0.6 ng/mL after 70 Gy, ~0.3 after 79 Gy, ≤0.1 after EBRT + BT boost.
- **No absolute nadir defines cure**. Meeting a definition of failure is for trials and reporting; **it does not by itself justify intervention**.

Feature	ASTRO 1996 (American Society for Therapeutic Radiology and Oncology)	Phoenix 2005 ("nadir + 2")
Definition of failure	Three consecutive PSA rises	PSA rise ≥2 ng/mL above the nadir
Date of failure	Backdated to midway between nadir and first rise	When nadir + 2 is reached, or when salvage starts if earlier
Weaknesses and strengths	Delays declaring failure by ≥18 months; backdating biases Kaplan–Meier estimates (worse with short follow-up); does not predict clinical progression or survival ; not developed for RT + ADT or BT	Best correlates with clinical local and distant relapse; valid with or without short-term ADT ; fewer false calls from bounces
Effect on reported results	Phoenix gives higher 5-year but lower 10-year biochemical recurrence-free survival than ASTRO	

- **Not failure**: PSA rises clearly due to laboratory error, **prostatitis responding to antibiotics**, or a **benign bounce** that resolves without intervention.

Significance of Time to Nadir

- Without ADT, PSA falls slowly after RT. Three sources contribute to the nadir: **residual benign epithelium, residual local cancer and subclinical micrometastases**. Higher dose ablates more benign epithelium, giving a lower nadir.

Scenario	Nadir	Time to nadir and later kinetics
No evidence of disease (NED) : benign epithelium only	Lowest	Longest time to nadir
Local failure	Intermediate	Steady fall until surviving cells outgrow dying ones, then a slow rise ; doubling time ≥1 year
Distant failure	Higher	Earlier nadir, then rapid rise; doubling time 3–6 months

- **Cut-offs**: nadir reached in **<12 months** → far more distant metastases at 5 years than a nadir after 12 months. **Biochemical failure within 18 months** of RT independently predicts distant metastasis and

prostate cancer mortality, with conventional 3D-CRT and with dose-escalated 76–78 Gy; failure after 18 months does not. In TROG 96.01, failure **before 24 months** predicted lower disease-specific survival.

Significance of Nadir Value and Doubling Time

- **Conventional-dose RT (~66–72 Gy):** median nadir **0.4–0.5 ng/mL with NED, >1.0 with local failure, >2 with distant failure.**
- **Nadir thresholds after EBRT:** nadir **>2 ng/mL at 1 year** predicts biochemical failure, distant metastasis and death. **>1.5 ng/mL at 2 years** is an independent risk factor for distant metastasis. **>0.5 ng/mL** predicts all-cause mortality (unfavourable risk, 70.2 Gy $\pm$ 6 months ADT).
- **After BT:** nadir **<0.5 ng/mL** → better long-term freedom from biochemical failure. PSA **<0.2 ng/mL at 5 years** → excellent 10-year control. PSA **$\leq$ 0.2 ng/mL at 4 years** → disease-free survival beyond 10 years in a very large multi-institutional analysis. With ^{125}I D90 of 160 Gy, only ~10% have PSA >0.2 at 5 years (two-thirds still falling), and the proportion below 0.2 keeps rising for **7–8 years**.
- **BT gives a deeper and later nadir than EBRT:** mean nadir about ten-fold lower after LDR or HDR + EBRT (~0.05–0.1 ng/mL) than EBRT alone (~0.5 ng/mL). Time to nadir is about **2.5 years after BT vs 1.5 years after EBRT.**
- **PSA doubling time (PSADT):** **<3 months** → associated with prostate cancer mortality. **<12 months** → markedly worse disease-specific survival (RTOG 92-02 analysis; in TROG 96.01 the HR for prostate cancer death was ~23). Practical use: consider **earlier intervention if PSADT <12 months**, and observation is more comfortable if it is longer.
- **Caveat:** PSADT is an imperfect surrogate. Its reliability depends on having many values over a wide range.

Brachytherapy and the Benign Bounce Phenomenon

- **Bounce (spike) = a rise >0.2 ng/mL followed by a durable decline.** It occurs after LDR and HDR BT and also after EBRT.
- Seen in about **a third of men after LDR**, and up to about **two-thirds of younger men**. The mean rise is ~1 ng/mL (up to **10 ng/mL**); it can last **12–18 months**, and **double peaks** occur.
- **A bounce is associated with better outcome:** improved biochemical control and survival compared with no bounce.
- A biopsy during a bounce may show residual cancer with treatment effect (indeterminate). **MR spectroscopy** shows a **diffuse** rise in choline:citrate across the gland, not the **focal** change expected with tumour. There is no reliable way to tell a bounce from failure in the first 3 years.
- **Management of a rising PSA after BT:** within **30 months** → monitor. If it **approaches 10 ng/mL** → systemic staging. If it **persists beyond 30 months** → prostate biopsy.

Biochemical Failure and Impact of Neoadjuvant Hormones

- With neoadjuvant ADT, PSA is often **already undetectable when RT starts**, so a true nadir and time to nadir **cannot be determined**.
- When testosterone recovers after EBRT + ADT, PSA often **bounces** (median ~2 years after treatment, ~0.8–0.9 ng/mL), then falls again as the RT effect continues. **Many more of these bounces meet the ASTRO definition of failure than the Phoenix definition.**
- Only about a fifth of bounces occur during testosterone recovery; the rest come **after testosterone has normalised**. This suggests a delayed testosterone response, **inflammation or instrumentation**. More frequent testing detects more bounces.
- **Phoenix (nadir + 2) makes fewer false calls of failure than ASTRO (three rises).**

Post-Radiation Therapy Imaging

GUIDANCE ON IMAGING AFTER RADIOTHERAPY

Body	Recommendation
AUA	mpMRI for a rising PSA after definitive treatment with a negative standard biopsy: a visible lesion improves biopsy yield over TRUS-guided biopsy alone. Routine follow-up: PSA only; no imaging without a PSA rise
American College of Radiology (ACR) Appropriateness Criteria	After RP with suspected recurrence: pelvic MRI with and without contrast is "usually appropriate". Biochemical recurrence after RT: bone scan first (a positive scan ends the workup); CT or MRI only if the bone scan is negative or inconclusive. No routine follow-up imaging
Society of Nuclear Medicine and Molecular Imaging	Appropriate-use criteria for PSMA PET to locate recurrence after biochemical failure

- Trials of what post-RT MRI changes mean, and whether it predicts local recurrence, are ongoing. **PSA remains the standard for monitoring.**

Post-Radiation Therapy Biopsy, Timing, and Interpretation

- **PSA cannot distinguish local from systemic failure**, so biopsy assesses the local effect. The problems are **timing, indeterminate biopsies with marked radiation effect and sampling error.**
- **Timing:** radiation causes **postmitotic cell death**, meaning fatally damaged cells survive a few divisions before dying. Biopsies taken as early as 6 months were once called failures. **Histological resolution runs parallel to the PSA nadir.** The **optimal biopsy time is 30–36 months**, although 24-month biopsies strongly predicted 7-year disease-free survival.
- **Grade only when radiation effect is absent or minimal.** Gleason grading was built on untreated surgical specimens and depends on gland architecture, which RT disrupts. Grading disintegrating glands gives a **false-positive, falsely high-grade** result and **unnecessary salvage.** **ADT-treated cancer can also mimic Gleason 4 or 5.**
- **Radiation atypia in benign glands can mimic cancer.** **High-molecular-weight cytokeratin** labels the **basal cell layer**, which is present in benign glands and absent in cancer.
- Radiation-effect scoring systems (Box 162.1) matter because they separate biopsies with **no or minimal effect from those with marked effect.** Ignoring the distinction dilutes the prognostic value of the biopsy and leads to inappropriate salvage. Proliferation markers (**proliferating cell nuclear antigen, Ki-67**) help judge residual tumour.
- **Prognostic value:** 2-year biopsy status predicts 10-year biochemical NED, freedom from distant metastasis and disease-specific survival. **Biopsies showing marked treatment effect behave as negative.** Biopsy status and Gleason score are the two strongest determinants of biochemical disease-free survival.

BOX 162.1 GRADING SCHEME FOR CYTOPLASMIC AND NUCLEAR RADIATION EFFECT

Component	Findings, in ascending severity
Cytoplasmic (0–3)	0: no radiation effect → 1: swelling, microvesicular change → 2: extensive vacuolation, macrovesicular change, voluminous cytoplasm → 3: indistinct or ruptured cytoplasm, or lipofuscin, or dilated glands or single cells with no gland formation
Nuclear (0–3)	0: no radiation effect → swelling or smudging with nucleoli still visible → smudged, distorted chromatin → nucleoli rare or absent → large bizarre nuclei → pyknotic small condensed nuclei
Total (0–6)	The two scores are added: 0–2 minimal, 3–4 moderate, 5–6 marked treatment effect

Imaging and Sampling Error

- **TRUS alone is of limited use after RT:** fibrosis changes the echogenicity of the irradiated gland.

- **MRI** far surpasses CT and ultrasound for soft tissue, and **mpMRI detects recurrence better than T2-weighted imaging alone**. Adding **MR spectroscopy** improves accuracy; **diffusion weighting** improves it; **dynamic contrast enhancement added nothing further** in one study.
- **Local recurrence arises at the site of the original dominant nodule**, as confirmed on whole-mount salvage specimens.
- MRI-TRUS fusion biopsy is well established, and **in-bore MRI-guided biopsy** is safe and effective.

KEY POINTS: EVALUATING THE RESPONSE TO RADIATION THERAPY

- Judge local therapy by **local eradication; lower and later nadir = better; bounces commoner after BT; viable tumour must be proven before radical salvage**, with mpMRI to guide biopsy.

Concept — After RT the gland stays in place, so every marker is ambiguous: PSA persists and bounces, and biopsies show dying cells. **Interpretation depends on time**: nadir kinetics, the 18-month failure interval, the 30-month bounce rule and the 30–36-month biopsy window.

TREATMENT EFFICACY

Outcomes for Conventionally Fractionated External Beam Radiotherapy

- **Higher dose → better tumour control**, a principle as old as the classic cell-survival (clonogenicity) curves. Early escalation studies used conventional **1.8–2 Gy** fractions.
- **Patterns of Care studies (1980s)** reviewed records from randomly selected U.S. departments: **higher doses to larger fields gave better control but more complications**. This spurred better planning and delivery and higher prostate doses.

TABLE 162.4 RANDOMIZED TRIALS OF CONVENTIONALLY FRACTIONATED STANDARD DOSE VERSUS DOSE-ESCALATED RADIATION THERAPY FOR PROSTATE CANCER

Trial	Risk mix; ADT	Standard vs escalated dose	Biochemical outcome (standard → escalated)
MD Anderson	All risks (mostly intermediate–high); no ADT	70 vs 78 Gy (conventional + 3D)	10-year freedom from failure 50% → 73%
PROG 95-09	Mostly low–intermediate; no ADT	70.2 vs 79.2 Gy-equivalent (photon + proton boost)	10-year biochemical failure 32% → 17% ; benefit in low and intermediate risk
Dutch trial	Mostly high; ~one-fifth given ADT	68 vs 78 Gy (3D-CRT)	7-year freedom from failure 45% → 56% ; benefit in intermediate and high risk
MRC RT01 (Medical Research Council)	All risks; all given ADT (3–6 months neoadjuvant and concurrent)	64 vs 74 Gy (3D-CRT)	5-year biochemical progression-free survival 60% → 71% ; benefit in all risk groups
GETUG 06	Mostly low–intermediate	70 vs 80 Gy (3D-CRT)	5-year biochemical failure-free survival 61% → 72%
RTOG 0126 (largest)	Low and intermediate	70.2 vs 79.2 Gy (3D-CRT or IMRT)	8-year biochemical failure-free survival 65% → 80% . Overall survival 75% vs 76% (HR 1.00) ; distant metastasis 6% → 4% (borderline, P = 0.05); less salvage therapy used

Table source: Kuban et al., 2008; Zietman et al., 2010; Al-Mamgani et al., 2008; Dearnaley et al., 2007; Beckendorf et al., 2011; Michalski et al., 2018.

- **What the table tests: every trial improved biochemical control, and none improved overall survival.** The benefit held across all risk groups, with or without ADT, which is **level 1 evidence making dose escalation the standard of care** (NCCN).
- **Why there was no survival gain:** effective, continually evolving **salvage therapy** for men who fail. In MD Anderson and RTOG 0126, the fall in distant metastasis only approached significance.
- **Trap:** these doses were delivered with 1990s–2000s techniques. Modern IMRT with image guidance means efficacy data must be read in light of the technique and dose used.
- **RT is a local treatment.** Its primary aim is local control, on the expectation that this prevents secondary dissemination.

Role of Radiation Therapy in High-Risk or Locally Advanced Disease

- Two trials randomised **ADT ± EBRT** in high-risk disease: **NCIC CTG PR.3/MRC PR07** (lifelong ADT by orchiectomy or LHRH agonist ± 65–69 Gy) and **SPCG-7/SFUO-3** (± a median 70 Gy); results are in the ADT trials table. Together they show that **RT improves overall survival by an absolute 8% at 7 years to 10% at 10 years** compared with ADT alone. Their doses were well below modern doses, which may be more effective.

KEY POINTS: TUMOUR CONTROL AFTER EXTERNAL BEAM RADIOTHERAPY

- **EBRT improves overall survival over ADT alone; ≥6 randomised trials → dose escalation is standard.**

Outcomes for External Beam Hypofractionation

- **Moderate hypofractionation = 2.25–4 Gy per fraction.**

TABLE 162.5 RANDOMIZED TRIALS OF MODERATE HYPOFRACTIONATION

Trial	Population	Conventional vs hypofractionated	Verdict
CHHiP	T1b–T3a N0; all 3–6 months neoadjuvant ADT	74 Gy/37 vs 60 Gy/20 vs 57 Gy/19	5-year failure-free survival 88.3% / 90.6% / 85.9%. 60 Gy/20 noninferior ; 57 Gy/19 not shown noninferior. No toxicity difference
RTOG 04-15	Low risk	73.8 Gy/41 vs 70 Gy/28	Noninferior disease-free survival (HR 0.85). More late grade 2–3 GI and GU toxicity
PROFIT	Intermediate risk	78 Gy/39 vs 60 Gy/20	Noninferior (HR 0.96); no difference in late grade ≥3 toxicity
HYPRO	Intermediate–high risk	78 Gy/39 vs 64.6 Gy/19 (3 days a week)	Not superior for relapse-free survival (HR 0.86)
NCIC trial	Mostly low risk	66 Gy/33 vs 52.5 Gy/20	Failure high in both arms, slightly worse with hypofractionation. Both doses low: designed before the α/β hypothesis, so the hypofractionated arm was biologically weaker at α/β 1.5
Early smaller trials	Mixed	2.4–3.1 Gy fractions, 55–72 Gy total	Hypofractionation equivalent or better; one showed more late grade ≥2 GU toxicity. Small numbers and low total doses

Table source: Dearnaley et al., 2012; Lee et al., 2016a; Catton et al., 2017; Incrocci et al., 2016; Lukka et al., 2005; Yeoh et al., 2011; Arcangeli et al., 2010; Pollack et al., 2011; Kuban et al., 2010.

- **What the table tests:** large trials show moderate hypofractionation **matches conventional fractionation, with a course half as long.** The standard schedule to know is **60 Gy in 20 fractions** (CHHiP, PROFIT).

- **Traps:** a trial designed for **noninferiority** (CHHiP, PROFIT, RTOG 04-15) differs from one designed for **superiority** (HYPRO, which was not superior). And **57 Gy/19 was not shown to be noninferior**.
- **Counselling:** acute (and in RTOG 04-15, late) GI and GU toxicity **may be higher** with larger fractions.
- **Postoperative hypofractionation:** a review of small retrospective and phase I/II series found acute grade ≥ 3 toxicity rare and late GU toxicity highly variable. The phase III trial of **HYPOR (62.5 Gy/25) vs COPORT (66.6 Gy/37)** showed hypofractionation **noninferior for patient-reported late GU and GI toxicity**; disease-control data are awaited.

Ultrahypofractionation

- **≥ 5 Gy per fraction = ultrahypofractionation (SBRT, SABR).** The evidence is mostly phase I/II series (Table 162.6) plus the phase III trials in the SBRT trials table above.
- **Pooled phase II analysis** (eight institutions): 5-year biochemical relapse-free survival **95% low, 84% intermediate, 81% high risk**; neither **total dose nor ADT** changed the outcome in any risk group.
- **Toxicity:** some series show clearly more late morbidity than moderate hypofractionation (grade 2 urinary toxicity $>30\%$ in some). Rare events: a **grade 4 rectal ulcer at 50 Gy/5** and a **rectal fistula**.
- **Dosimetric risk factors for high-grade rectal toxicity:** rectal wall volume receiving **50 Gy**, and **$>35\%$ of the rectal circumference receiving >39 Gy**.
- **Hydrogel rectal spacer at 45 Gy/5** (phase II): periprostatic rectal ulcers fell from an expected near-universal rate to a small minority, with **no late grade ≥ 3 GI toxicity**.

TABLE 162.6 SELECTED SERIES OF ULTRAHYPOFRACTIONATION

Dose (5 fractions)	Populations	Biochemical control	Late toxicity
35–37.5 Gy (most series)	Mostly low–intermediate; some high risk, some ADT	~ 94 – 100% at 2–7 years in low and intermediate risk; lower in high risk ($\sim 77\%$ at 3 years)	Grade ≥ 2 GU 4–31% ; grade ≥ 2 GI 1–9%; erectile dysfunction 19–43% where reported
40 Gy	Low–intermediate	5-year disease-free survival $\sim 97\%$	Grade 3 GU $\sim 1\%$; GI 0%
45–50 Gy (escalation)	Low–intermediate	5-year freedom from biochemical failure $\sim 99\%$	Grade ≥ 3 GU $\sim 5.5\%$, GI $\sim 7\%$

Data from Katz and Kang, 2014 (series: Friedland 2009; King 2012; Bolzicco 2013; Chen 2013; Loblaw 2013; Oliai 2013; Katz and Kang 2014; Hannan 2016; Meier 2018).

- **What the table tests:** biochemical control is excellent at every dose level, so **escalating beyond ~ 36 – 40 Gy buys toxicity, not control**. This matches the meta-analysis in the SBRT section.

KEY POINTS: HYPOFRACTIONATION AND STEREOTACTIC BODY RADIATION THERAPY

- Requires **accurate set-up and conformal planning**. Moderate hypofractionation (intact and postoperative): **noninferior control and toxicity to 5 years**. Ultrahypofractionation: good early control, variable toxicity, phase III follow-up ongoing.

Concept — Dose escalation improved biochemical control but not survival. Hypofractionation then delivered equal control in fewer visits by exploiting the low α/β . In both, the limiting factor is **rectal and urethral tolerance, not tumour control**.

OUTCOMES FOR BRACHYTHERAPY

- **Post-implant biopsies** from early series showed a **very low rate of positive biopsies at 2 years** after implants with good dosimetry. Biopsy has limits, as discussed above, so biochemical surrogates and survival are now used.

- **PSA is the most useful marker of relapse:** a rising PSA predicts local and/or distant relapse years before clinical recurrence. It is hard to interpret because of **variability at low levels** and **suppression by even short ADT**. A **bounce** occurs in about a third of patients after BT and has been linked to **better** cancer control.

COMPARATIVE EFFICACY OF EXTERNAL BEAM RADIATION THERAPY VERSUS BRACHYTHERAPY

- **No randomised trial has compared BT alone with EBRT alone.** The evidence is retrospective.
- **Intermediate risk:** ^{125}I BT (145 Gy) gave better long-term freedom from biochemical failure than EBRT (~75 Gy), with **no difference in distant metastasis, disease-specific or overall survival**. BT remained independent after propensity matching, although BT patients were younger, had less aggressive disease and received less ADT.
- **cT1–T2, all risks:** only **EBRT <72 Gy** was inferior. Otherwise the modality did not predict outcome, and RP and EBRT + BT were equivalent.
- **HDR monotherapy** for low-risk disease shows acceptable toxicity and early biochemical control similar to permanent seeds.

TABLE 162.7 SELECTED SERIES COMPARING FREEDOM FROM BIOCHEMICAL RELAPSE: IMPLANT VERSUS EXTERNAL BEAM RADIATION THERAPY

Comparison	Risk group	Direction of result
LDR ^{125}I monotherapy (≥ 144 –160 Gy) vs EBRT 70–80 Gy (3D-CRT or IMRT)	Low and favourable intermediate	BT better biochemical relapse-free survival ; in one series significant only at lower CAPRA scores
LDR boost (110 Gy) + IMRT 40–45 Gy vs IMRT 72–76 Gy	Unfavourable intermediate and high	BT combination better ; the gap widens with a stricter PSA >0.2 ng/mL failure criterion
EBRT + HDR boost (10–20 Gy) vs EBRT 70–74 Gy	Intermediate	HDR boost better biochemical failure-free survival

Table source: Delouya et al., 2017; Smith et al., 2015; Tanaka et al., 2017; Tsubokura et al., 2018. All retrospective same-institution series.

- **Pattern and trap:** every series favours BT, but all are **retrospective and biochemical-only**, and the EBRT arms used doses below modern standards. Biochemical superiority has **not** been shown to translate into survival.

TABLE 162.8 HIGH-DOSE-RATE BRACHYTHERAPY AS MONOTHERAPY

Schedule	Example doses	Biochemical disease-free survival
3 fractions	30–34.5 Gy	89–94%
4 fractions	38 Gy (the most frequent schedule)	91–97%
6–9 fractions	42–54 Gy	93–97%

Median follow-up 40–83 months. Table source: Demanes et al., 2011; Shah et al., 2012; Barkati et al., 2012; Zamboglou et al., 2013; Ghadjar et al., 2014; Potharaju et al., 2015; Yamazaki et al., 2018.

- **What the table tests:** multifraction HDR monotherapy consistently gives about **90–97% biochemical control**. All schedules use **≥ 3 fractions**, in contrast with the failed single 19 Gy fraction.

BRACHYTHERAPY COMBINED WITH EXTERNAL IRRADIATION

- Here BT acts as a **boost** to escalate dose within the prostate and periprostatic tissue. Control rates are given in the brachytherapy section above.

- **RTOG 0232 design:** intermediate risk (cT1c–T2b by the 6th edition of the American Joint Committee on Cancer [AJCC] manual, with **Gleason 2–6 and PSA ≥ 10 but < 20 , or Gleason 7 and PSA < 10**). BT alone (^{103}Pd 125 Gy or ^{125}I 145 Gy) vs partial-pelvis EBRT 45 Gy followed 2–4 weeks later by BT (^{103}Pd 100 Gy or ^{125}I 110 Gy). **Limitations:** not stratified into favourable and unfavourable intermediate risk, and ADT was at investigator discretion.
- **ASCENDE-RT design:** stratified by NCCN risk group. All patients had **12 months of ADT starting 8 months before RT**, then EBRT 46 Gy to the prostate, seminal vesicles and regional nodes. They were then randomised to a further **32 Gy EBRT prostate boost** or a ^{125}I implant to **115 Gy** 2–3 weeks after EBRT.
- **Toxicity:** several (not all) studies show **more GI and GU toxicity** with combined BT + EBRT than with either alone. Patients must be **counselled that the control benefit is partly offset by morbidity**.

Concept — *BT's dose advantage shows up as better biochemical control. For low and intermediate risk, BT alone is enough. Adding EBRT improves control only where disease may extend beyond the prostate, and always at a toxicity cost.*

OUTCOMES OF MODERN TECHNIQUE

Treatment Morbidity and Quality-of-Life Outcomes

- **70 Gy was long considered the maximum safe dose.** Conformal RT made higher doses possible, but **toxicity tracks dose to the organ**. In the MD Anderson trial, 5-year grade ≥ 2 complications were about **twice as common at 78 Gy as at 70 Gy**. Rectal complication risk rose substantially when **more than 25% of the rectum received 70 Gy**.
- So technology that cuts organ dose, such as IMRT, reduces toxicity.

TABLES 162.9 AND 162.10 ACUTE AND LONG-TERM TOXICITY IN PATIENTS TREATED WITH THREE-DIMENSIONAL CONFORMAL RADIATION THERAPY VERSUS INTENSITY-MODULATED RADIATION THERAPY

Toxicity	Acute (Table 162.9)	Late (Table 162.10)
GI	Lower with IMRT in most series, despite equal or higher doses (one series at 81 Gy showed a very low rate in both arms)	Lower with IMRT in most series; the reverse in one series (see the flagged discrepancy)
GU	Mixed: lower with IMRT in some series, higher at 81 Gy in one, no difference in others	No consistent benefit; higher with IMRT in the 81 Gy series; otherwise not significant

Toxicity scales: NCI Common Terminology Criteria for Adverse Events (CTCAE), RTOG/EORTC, and the Fox Chase modified Late Effects Normal Tissue Task Force scale (LENT). Table source: Zelefsky et al., 2008a; Al-Mamgani et al., 2009; Sharma et al., 2011; Michalski et al., 2013; Dolezel et al., 2015; Alongi et al., 2009; Wortel et al., 2015, 2016.

- **What the tables test: IMRT reduces GI but not GU toxicity** compared with 3D-CRT, while delivering higher prostate doses. The rectum lies outside the target and can be spared; the urethra lies **inside** it and cannot.

Urinary Toxicity and Management

- **IMRT to high dose (up to 86.4 Gy):** 10-year grade ≥ 2 urinary toxicity ~20%; grade 3 rare. **Image guidance halved late grade ≥ 2 GU toxicity** (3-year rates ~10% vs ~20%).
- **Patient-reported symptoms:** dysuria, weak stream and frequency **peak at about 2 months**, improve by 6 months and are **back to baseline by 12 months**. **Incontinence is uncommon after RT**.
- **RTOG 94-08 (intact prostate):** acute grade ≥ 2 / ≥ 3 urinary toxicity **30.8% / 3.8%**; late at 5 years **18.3% / 3.3%**. After prostatectomy (**RTOG 96-01**), acute grade ≥ 2 was lower (14%) and the other rates were similar.

- **After BT:** IPSS rises immediately and falls at a rate set by the isotope's half-life. Chronic grade ≥ 2 urinary toxicity after permanent implants is $\sim 20\%$. **α -blockers started before implantation** may reduce the severity and duration of symptoms.
- **Predictors after LDR:** the length of urethra receiving high dose (grade 2–3 morbidity when about 20 ± 11 mm received >400 Gy with ^{125}I); gland volume $>60 \text{ cm}^3$; and pretreatment IPSS. IPSS >20 carries $\sim 29\%$ retention risk; IPSS <10 only $\sim 2\%$.
- **HDR:** a similar pattern, with acute grade ≥ 2 urinary symptoms in **15–18%** and chronic grade ≥ 3 in **5–10%**.
- **ASCENDE-RT:** 5-year grade 3 urinary toxicity **18.4% with the LDR boost vs 5.2% with the EBRT boost** ($P < 0.001$). This is the price of the halved biochemical failure.
- **Incontinence after TURP:** higher risk after BT with a previous TURP, and higher still after **multiple TURPs**. Without urethral trauma it stays at $\sim 3\%$.

Rectal Toxicity and Management

- **Acute rectal symptoms are self-limiting.** Late radiation proctitis with bleeding usually appears **6–18 months** after treatment.
- **Management ladder:** steroid suppositories or sucralfate enemas (for minor, self-limited cases) $\rightarrow$ argon plasma coagulation (APC) or hyperbaric oxygen $\rightarrow$ diverting colostomy in the worst cases. APC controls bleeding in most patients but can cause ulceration and, rarely, colovesical fistula.
- **IMRT to 81 Gy:** 10-year grade ≥ 2 GI toxicity $\sim 5\%$, grade 3 $<1\%$. Low with or without fiducial-based image guidance.
- **Patient-reported bowel symptoms:** urgency and frequency **peak at 2 months** and partly recover.
- **ProtecT:** RT (3D-CRT) caused more **bowel bother** than active monitoring or RP, worst at **6 months** and improving thereafter, with a residual increase in **bloody stools at 2–6 years**. **No difference in loose stools or faecal incontinence.**
- **BT:** significant rectal complications in **1–2%** and bleeding in $\sim 1\%$, **lower than after EBRT**. **Recto-urethral fistula is rare but feared;** a colostomy rate of $\sim 0.3\%$ was found in Medicare claims.
- **Rectal dose limits for BT:** complications are rare when the rectal wall receives $\leq 85\%$ of the prescription. For BT alone, rectal V100 $<1.0 \text{ cm}^3$ on day-1 dosimetry or $<1.3 \text{ cm}^3$ on day-30 dosimetry (ABS). For BT + EBRT, bleeding risk rises with rectal V100 $>0.5 \text{ cm}^3$ and EBRT rectal V30 $>35\%$.

Rectal-Sparing Strategies

- **Options:** precise MRI delineation of the anorectum; **endorectal balloon;** a saline-filled balloon between prostate and rectum; and, most widely reported, a **transperineal spacer** (biodegradable **hydrogel or hyaluronic acid**) placed in **Denonvilliers' fascia** to separate the prostate from the anterior rectal wall (Fig. 162.9 — axial T2 MRI showing hydrogel between the prostate/seminal vesicles and the rectum).
- **Safe with LDR, HDR, EBRT, SBRT and combined therapy.** It can be placed **before treatment or after suboptimal dosimetry from a permanent implant.**
- **Hydrogel randomised trial** (IG-IMRT 79.2 Gy): late rectal toxicity **2.0% vs 7.0%** ($P = 0.04$).
- **Hyaluronic acid phase III trial** (60 Gy/20; 2:1; spacer + fiducials vs fiducials alone): grade ≥ 2 rectal toxicity **2.9% vs 13.8%.**

Erectile Dysfunction and Management

- The structure responsible for radiation-associated erectile dysfunction (ED) is **unknown**. The mechanism is **predominantly vascular**, and **penile bulb dose** may correlate.

- Poor erections rise over **24 months** after RT, mostly in patients who also received **ADT**; the decline is modest with RT alone. After SBRT, about half of potent men remain potent at 5 years. Predictors: **baseline health-related quality of life and age** at 2 years, then quality of life and **body mass index** at 5 years.
- **Potency preservation by modality**, best to worst: **BT alone** (~76% at 5 years) > **EBRT + BT** > **EBRT + BT + ADT** (~52–58%). BT alone also beats prostatectomy and EBRT. **Adding EBRT and diabetes mellitus** (multivariate) and **age** (univariate) predict loss of potency.
- **Phosphodiesterase type 5 inhibitors (PDE5Is)** are effective in about **three-quarters** of men who use them. After SBRT, sexual aids still worked at 5 years in most men who had used them before RT, and in most who had not.
- Men treated with RT were **less likely than prostatectomy patients to try non-PDE5I aids** (urethral alprostadil [MUSE], injections, vacuum device). It is unclear whether this reflects referral practice or less dysfunction.

KEY POINTS: TREATMENT MORBIDITY AND QUALITY-OF-LIFE OUTCOMES

- Rectal bleeding mostly mild; spacers reduce proctitis; incontinence uncommon; PDE5Is usually effective.

Concept — Toxicity depends on the dose each organ receives, not the dose to the prostate. Organs **outside** the target (rectum) benefit from IMRT, IGRT and spacers. The organ **inside** it (urethra) does not, so GU toxicity is the persistent limit. ED is **vascular and driven by ADT**, and BT alone preserves potency best.

SALVAGE TREATMENT IN THE EVENT OF LOCAL FAILURE

- **Prerequisite: confirmed local-only failure** (confined to the prostate). Options: salvage prostatectomy, repeat EBRT, BT, cryotherapy, high-intensity focused ultrasound and irreversible electroporation.

Option	Evidence	Control	Principal toxicity
Salvage prostatectomy	CALGB 9687 (Alliance): biopsy-proven recurrence after ≥60 Gy for cT1–2 disease	~39% salvaged at 5 years (other series 30–80%)	Incontinence ~63% at 2 years (series 30–70%); rectal and obturator nerve injuries
EBRT re-irradiation (SBRT)	Multicentre series, biopsy-proven local recurrence	About half salvaged at 3–5 years	Grade ≥2 GU ~one in five; grade ≥3 GU ~8%; GI minimal
LDR BT	Series; NRG/RTOG 0526 (phase II, partial or whole gland)	4-year PSA control 40–90%; NRG 0526: 68% salvaged at 5 years	NRG 0526: late grade 3 GI/GU ~14%, no grade >3 . Other series: grade 3–4 toxicity, fistula, colostomy or urostomy
HDR BT	MSKCC phase II (8 Gy × 4 in one insertion over 2 days); large retrospective series	~70% biochemical relapse-free at 5 years	Urinary: grade 2 events, strictures needing dilation and acute obstruction needing a catheter; rectal toxicity minor

- **Why EBRT re-irradiation is hard**: the bladder, rectum and bowel are already near tolerance, the **urethra lies within the original target**, and modern dose escalation used the highest dose upfront.
- **Why BT is favoured**: sources inside the prostate give highly conformal dose, **minimising bladder, urethral and rectal dose** while delivering high dose to tumour. Both LDR and HDR are used.
- Ongoing: **focal HDR** in one or two fractions to the MRI-defined lesion with a small margin.

Concept — Salvage is limited by **the dose normal tissue has already received**. BT is preferred because it concentrates new dose inside the prostate, and **biopsy-proven local-only failure** is the prerequisite for any radical salvage.